

1. System Overview

The system consists of multiple interconnected components.

1.1 Architecture

The architecture follows a microservices pattern.

1.1.1 Frontend

The frontend is built with React and TypeScript.

It communicates with the backend via REST API.

1.1.2 Backend

The backend is a Python FastAPI application.

1.1.2.1 API Layer

The API layer handles HTTP requests and authentication.

1.1.2.2 Service Layer

The service layer contains business logic and data access.

Database operations are abstracted through a repository pattern.

1.2 Deployment

The system is deployed on Kubernetes with auto-scaling.

2. Data Pipeline

Data flows through several processing stages.

2.1 Ingestion

Raw data is ingested from multiple external sources.

2.2 Transformation

Data is cleaned, normalized, and enriched with metadata.

2.2.1 Cleaning

Invalid records are filtered out using validation rules.

2.2.2 Normalization

Values are scaled to the [0, 1] range using min-max scaling.

2.3 Storage

Processed data is stored in PostgreSQL with indexing.

3. Results

The system processes 1 million records per hour.

3.1 Performance

Average latency is 50ms per request at the 95th percentile.

3.2 Accuracy

Classification accuracy is 95.3 percent on the test set.